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| | S.S.V.P.S's B. S. Deore Polytechnic, Dhule<br>**Department of Computer Engineering**<br>*Academic Year : 2025 – 2026* |

# Presentation on Capstone Project Execution (CPE–316004)

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| **Topic** | Web Application for Water Jar Distribution System | |
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| | 3. Tanishka Hemant Patil | |
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| **Guided by** | Mr. C.P.Bhamare | |
| **Semester** | **Sixth (K Scheme)** | **Date : 17/4/2026** |

# Abstract

The proposed system “Web Application for Water Jar Distribution System” is developed to help a water jar supplier manage customer orders, deliveries, and payment records efficiently. Currently, the supplier maintains records manually, which may lead to errors and difficulty in tracking orders.

This system stores all customer and order information digitally and allows customers to place orders online, while the supplier can easily manage deliveries and records.

# Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

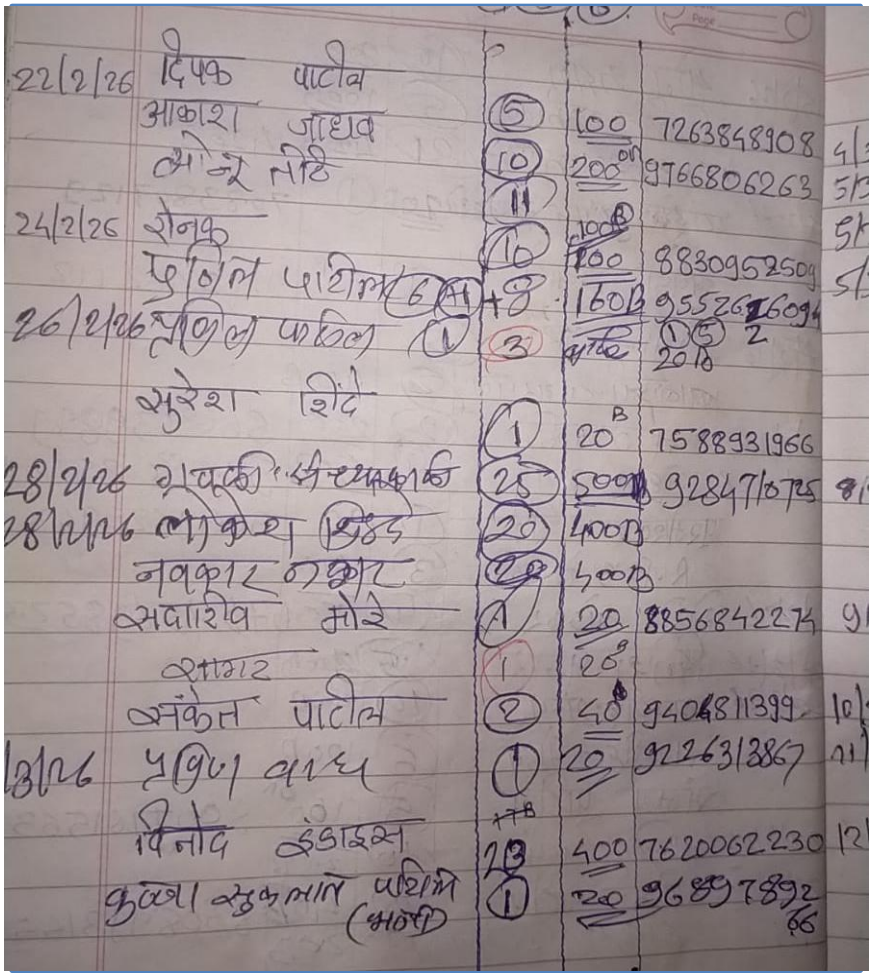
# 1. Introduction

- Drinking water is a basic need for homes, offices, and workplaces, and many people use water jars for safe drinking water.
- Water jar supply is a common business where a supplier delivers filled jars and collects empty jars from customers regularly.
- Customers place orders for water jars whenever they need them for daily use.
- In existing business, the supplier manages orders, deliveries, and payments manually using notebooks or phone calls.
- Because of manual records, it becomes difficult to track customer details, jar deliveries, and payment status, which may cause mistakes or missing records.
- Therefore, maintaining proper records of customers, orders, deliveries, and payments is important for managing the water jar supply business efficiently.

# 2. Literature Review

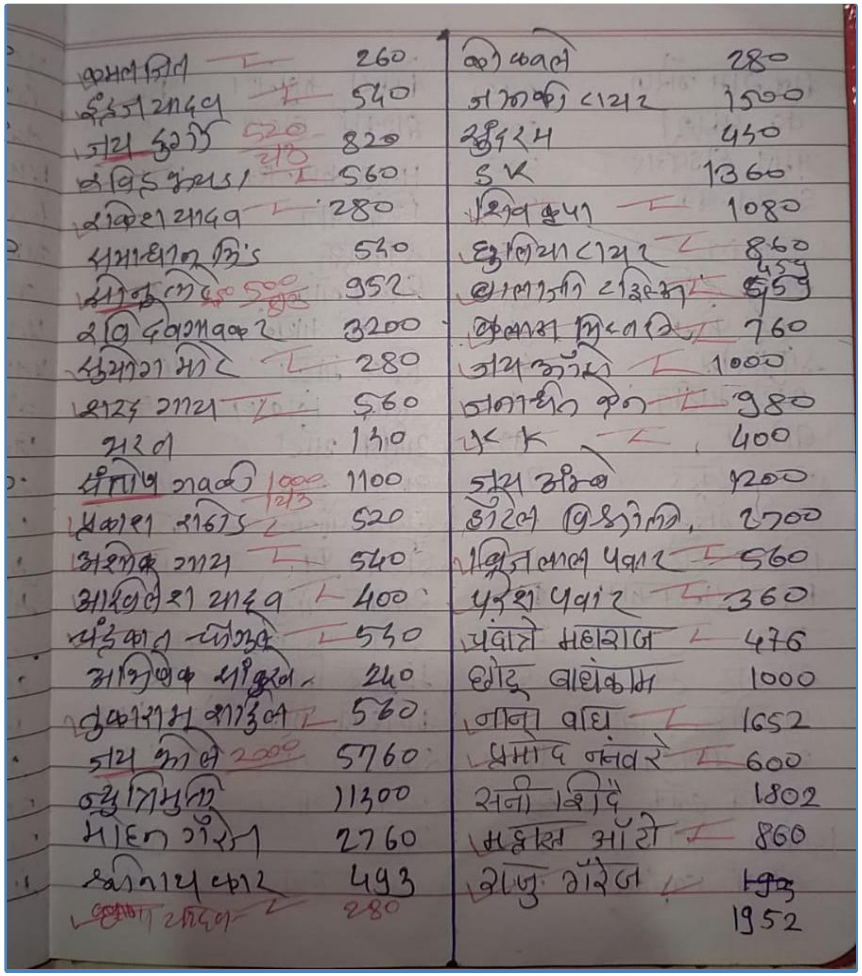
- Currently, the supplier manages all records manually.
- Customer details, jar orders, and delivery information are written in a notebook.
- The supplier records the number of water jars delivered and collected from customers.
- Payment details of customers are also maintained manually in the same notebook.
- Because all information is written by hand, it becomes difficult to check previous records quickly.
- Sometimes mistakes or missing records may occur in orders or payments.
- Managing daily or monthly jar delivery records takes more time and effort for the supplier.

# 2. 1 Study of existing system



- The supplier maintains order records manually in a notebook.
- The notebook contains order date, customer name, and number of water jars ordered.
- The supplier also writes the total payment amount based on the number of jars ordered.
- Each customer's phone number is recorded for contact and delivery purposes.
- All calculations and records are written by hand, which can make tracking and managing data difficult.

# 2.1 Study of existing system



- The supplier records regular customer names and total payment in a notebook.
- It shows the weekly or monthly amount of each customer.
- A red tick mark indicates payment is completed.
- All records are maintained manually.

# 2.2 Problem Statement

**Existing System / Manual Process**

- Currently, the water jar supplier manages customer details, orders, deliveries, and payments manually.
- Orders are taken through phone calls or in person and recorded in notebooks.
- Delivery details, empty jar returns, and payment records are also written manually.

# 2.2 Problem Statement

**Problems and Limitations**

- Order details may be missed or written incorrectly in the notebook.
- It is difficult to track daily, weekly, and monthly jar deliveries.
- There is no proper record of pending deliveries and empty jar returns.
- Managing customer payment details manually can be confusing.
- Manual record keeping takes more time and effort.

**Need for Improvement**

- A digital water jar management system is needed to store customer, order, delivery, and payment data in one place.
- This will reduce manual work and make the process faster and more organized.

# 2.3 Problem Definition

The proposed system is a web-based application designed to manage water jar orders, deliveries, and payments easily. In the current system, the supplier maintains records manually using notebooks or phone calls, which makes it difficult to track orders, deliveries, and payments. Therefore, an online system is needed to manage water jar supply work in a simple and organized way.

# 3. Scope of the project

- The system allows customers to place water jar orders online.
- It helps the supplier store and manage customer details easily.
- The supplier can view and manage all customer orders in one place.
- The system allows assigning orders to delivery staff.
- The supplier can update the delivery status of each order.
- It helps in managing payment status (received or pending).
- The system also allows the supplier to view daily water jar orders.

# 3.1 User Requirements

**Customer Requirements**

- Customer should be able to place water jar orders online.
- Customer should enter name, contact number, address, and jar quantity.
- Customer should be able to view order details.
- Customer should get order confirmation.

# 3.1 User Requirements

**Supplier (Admin) Requirements**

- Supplier should be able to view all customer orders.
- Supplier should manage customer details.
- Supplier should update delivery status.
- Supplier should manage payment records.
- Supplier should view daily, weekly, and monthly order details.
- Supplier should be able to search customer records easily.

# 3.1 User Requirements

**Delivery Staff Requirements**

- Delivery staff should be able to log in to the system.
- Delivery staff should be able to view only the orders assigned to them.
- Delivery staff should see customer name, address, and jar quantity.
- Delivery staff should update delivery status after delivery.
- Delivery staff should check their daily delivery list.

# 3.2 System Requirements

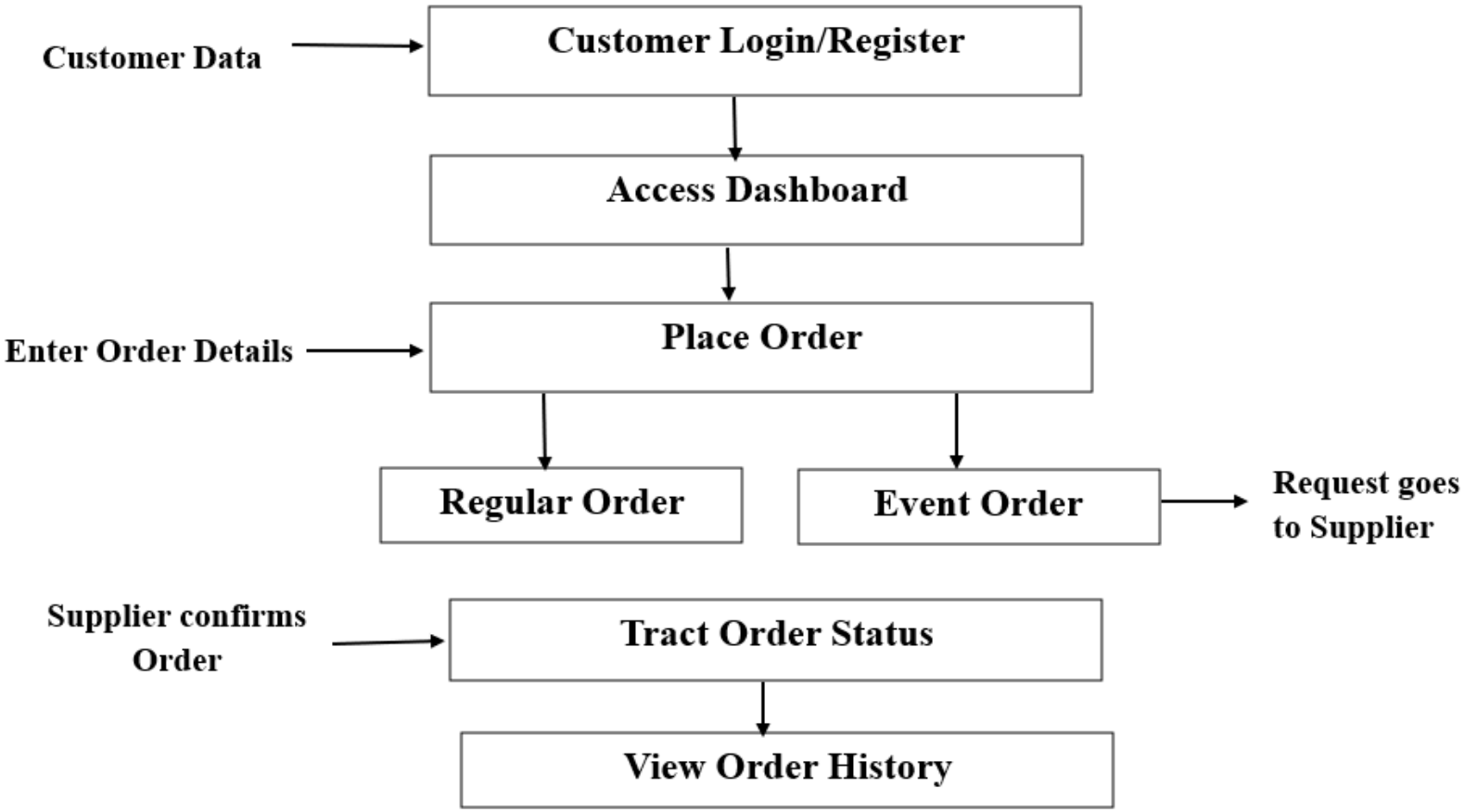
- Programming Languages - JavaScript
- Frontend - React JS, HTML, CSS
- Backend - Node.js, Express.js
- Database - MongoDB
- Development Tools - VS Code
- Operating System – Windows 10

# 4. Methodology

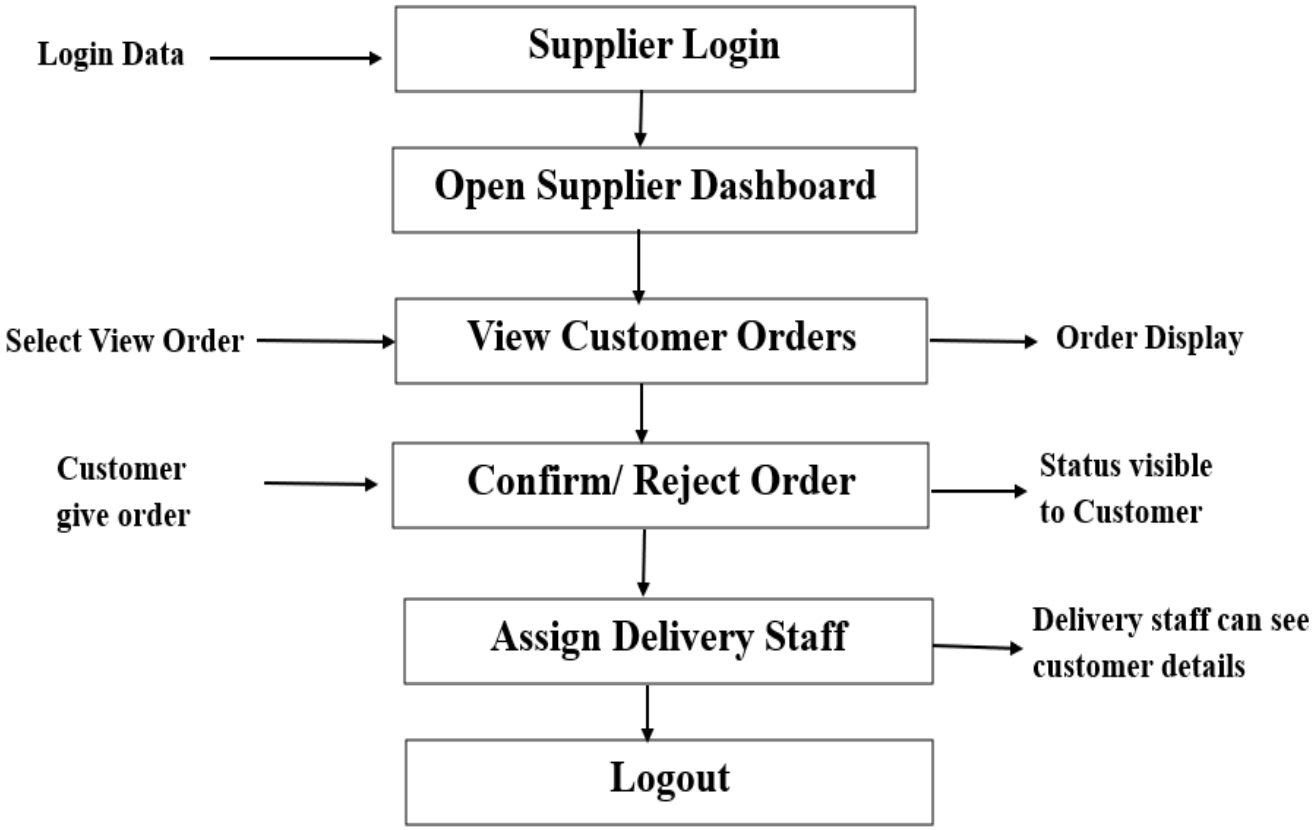
**Modules**

- Customer Module
- Supplier Module
- Delivery Staff Module

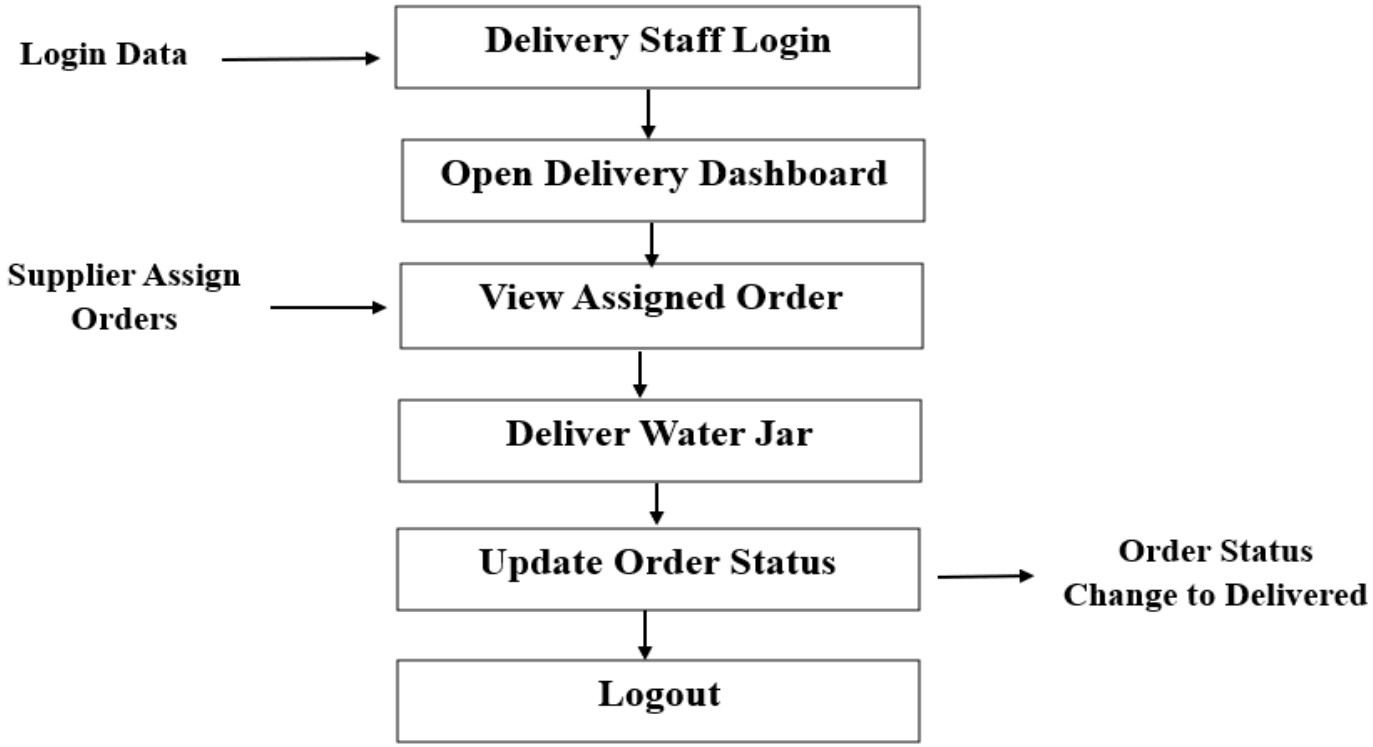
# 4. 1 Customer Module



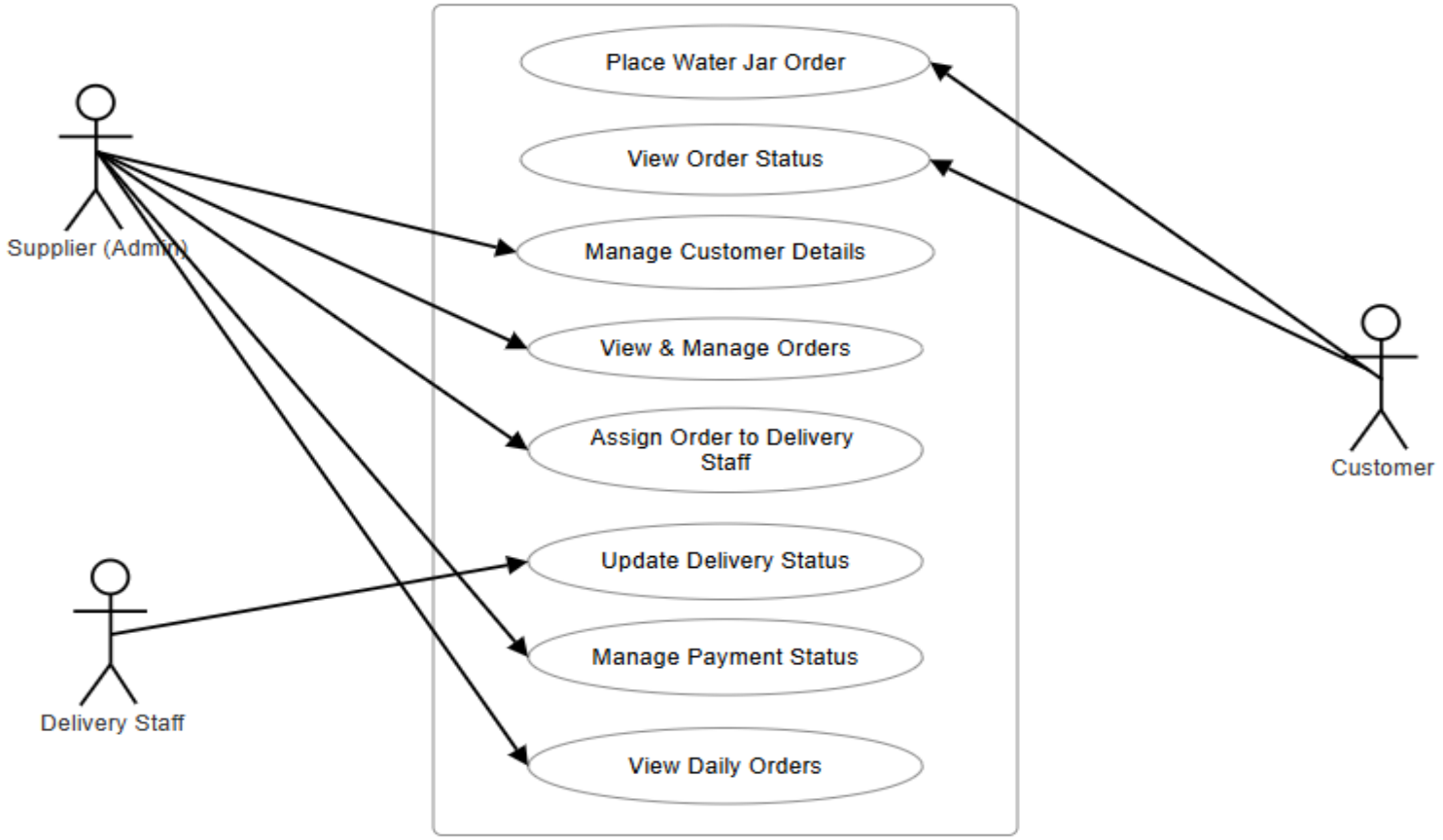
# 4. 2 Supplier Module



# 4. 3 Delivery Module



# 4. 4 Use Case Diagram



# 4. 5 Tools and Technology

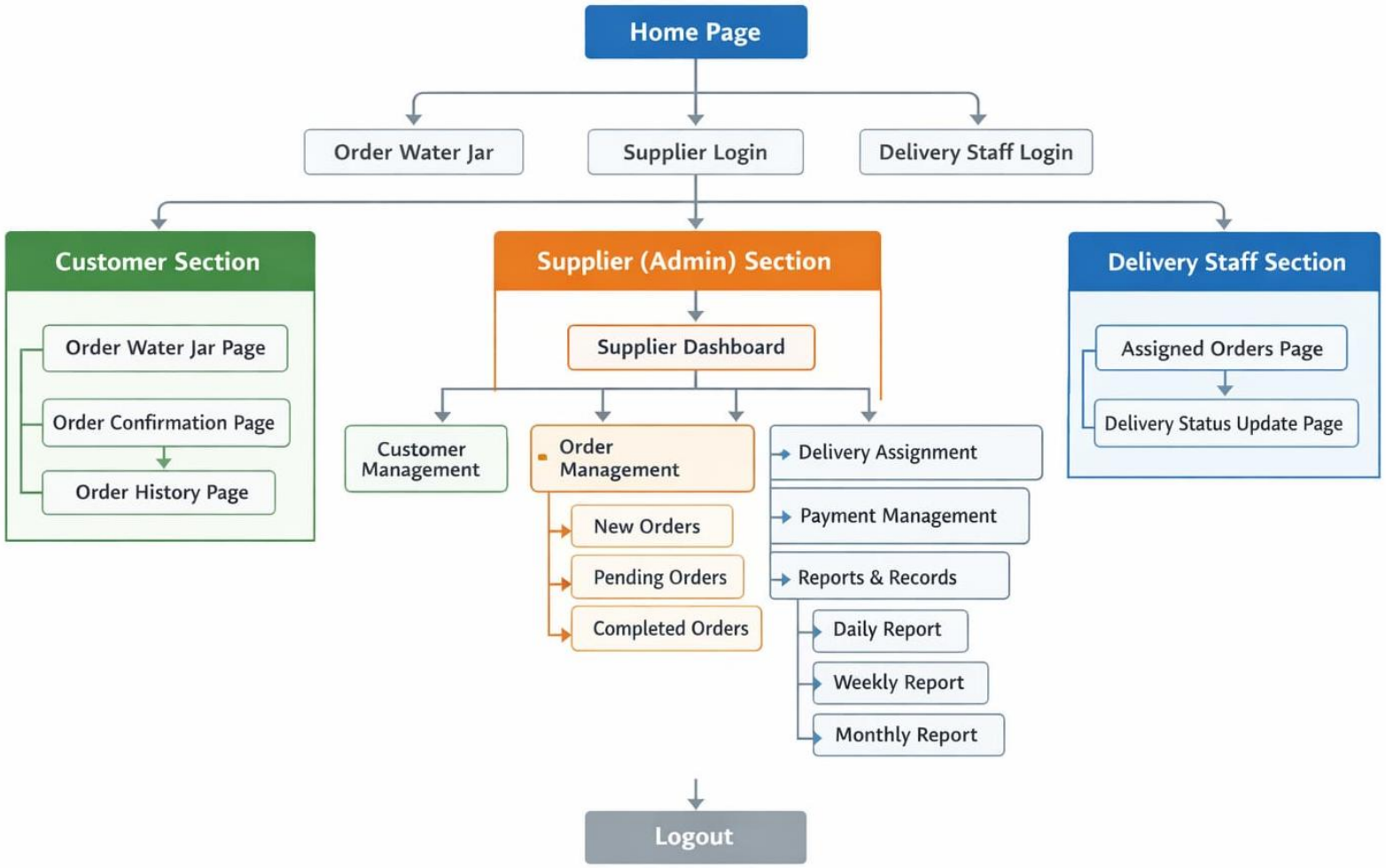
- Programming Languages - JavaScript
- Frontend - React JS, HTML, CSS
- Backend - Node.js, Express.js
- Database – MongoDB
- Development Tools - VS Code
- Operating System – Windows 10

# 5. Details of designs, working & processes

**Flow of Operations**

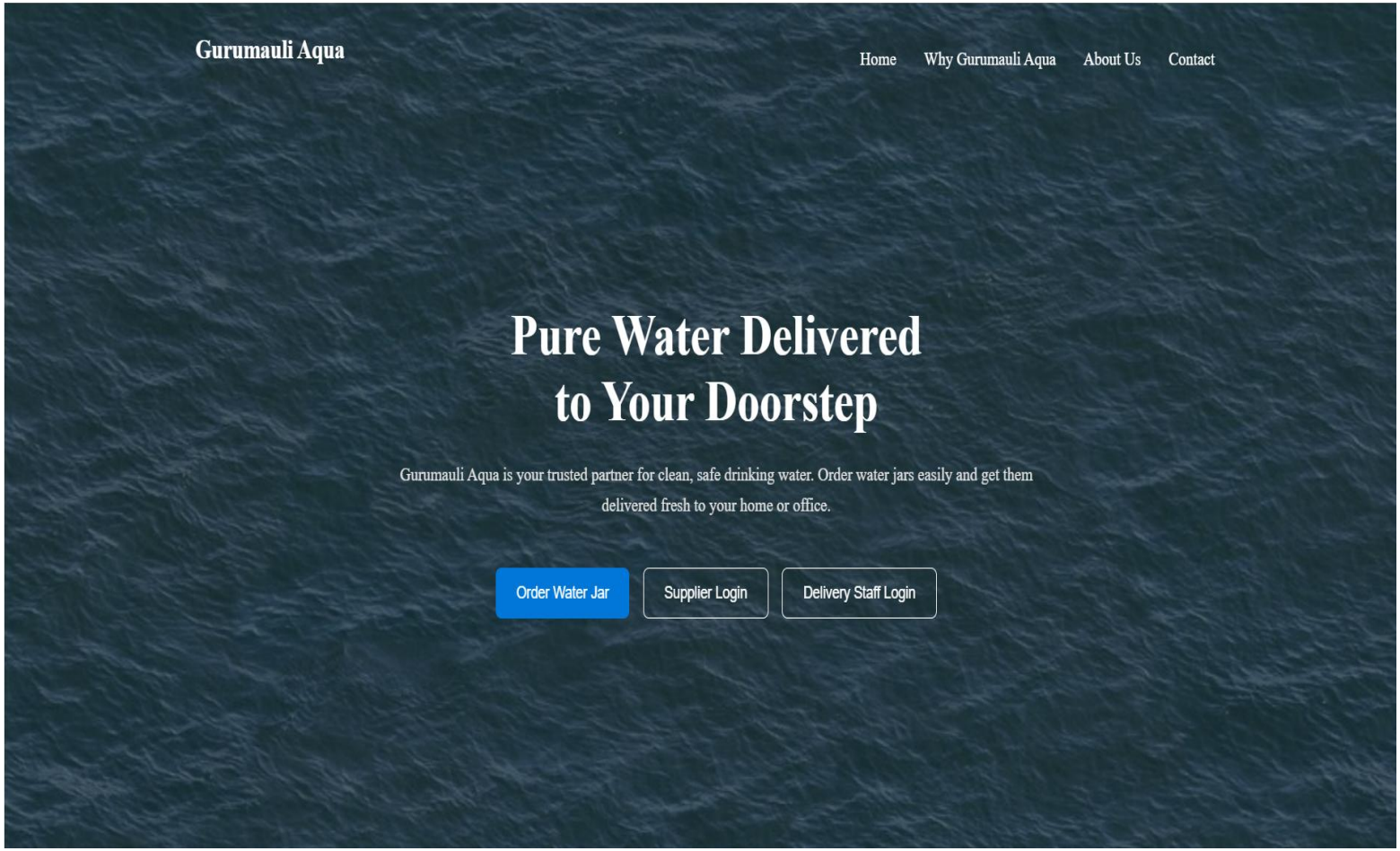
1. Customer places a water jar order
2. Order is stored in the system
3. Supplier views the order
4. Supplier assigns order to delivery staff
5. Delivery staff delivers jars
6. Delivery status and payment are updated
7. Supplier views report

# 5. Details of designs, working & processes



# 6. Reults and Applications

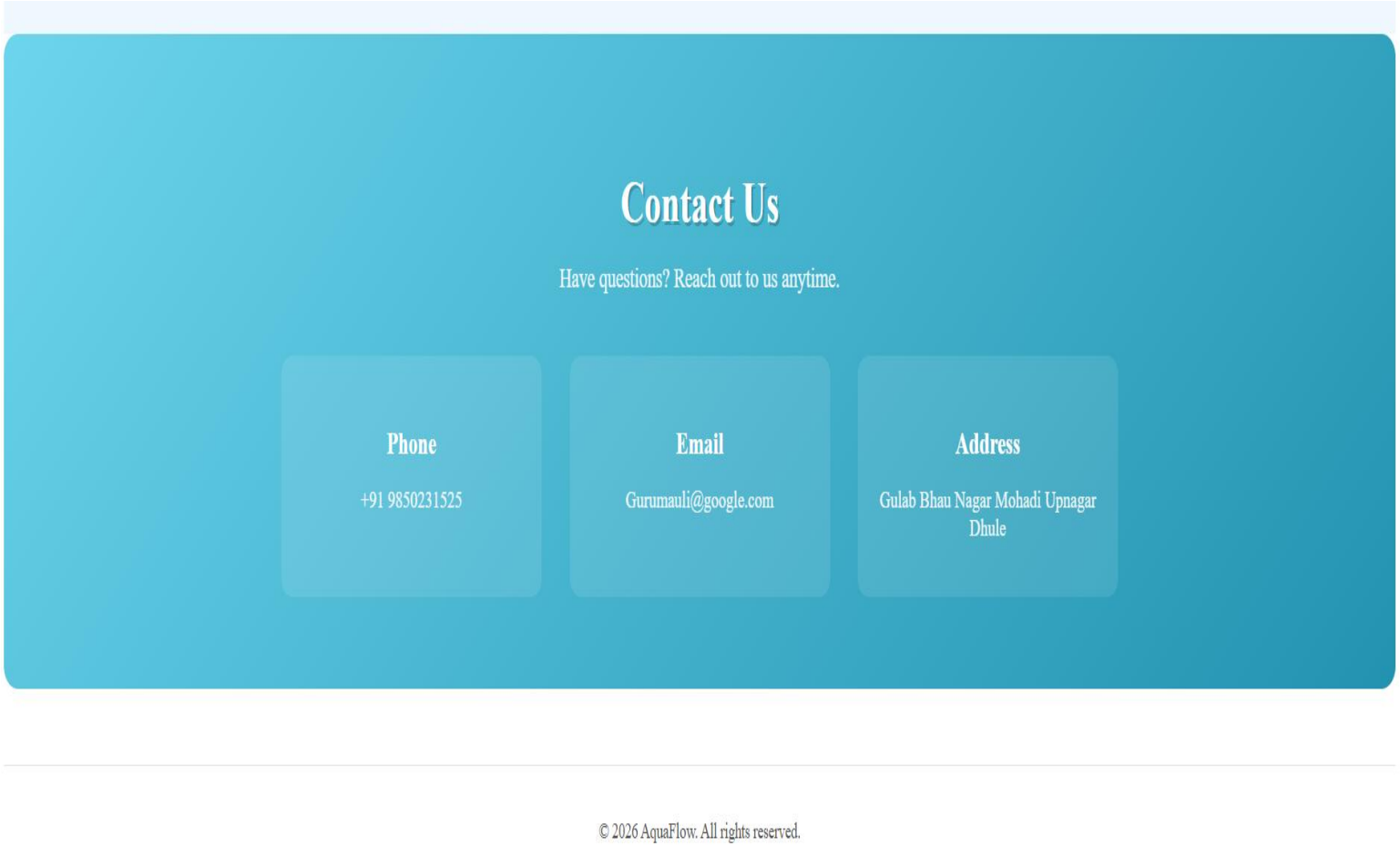
- **Home Page**



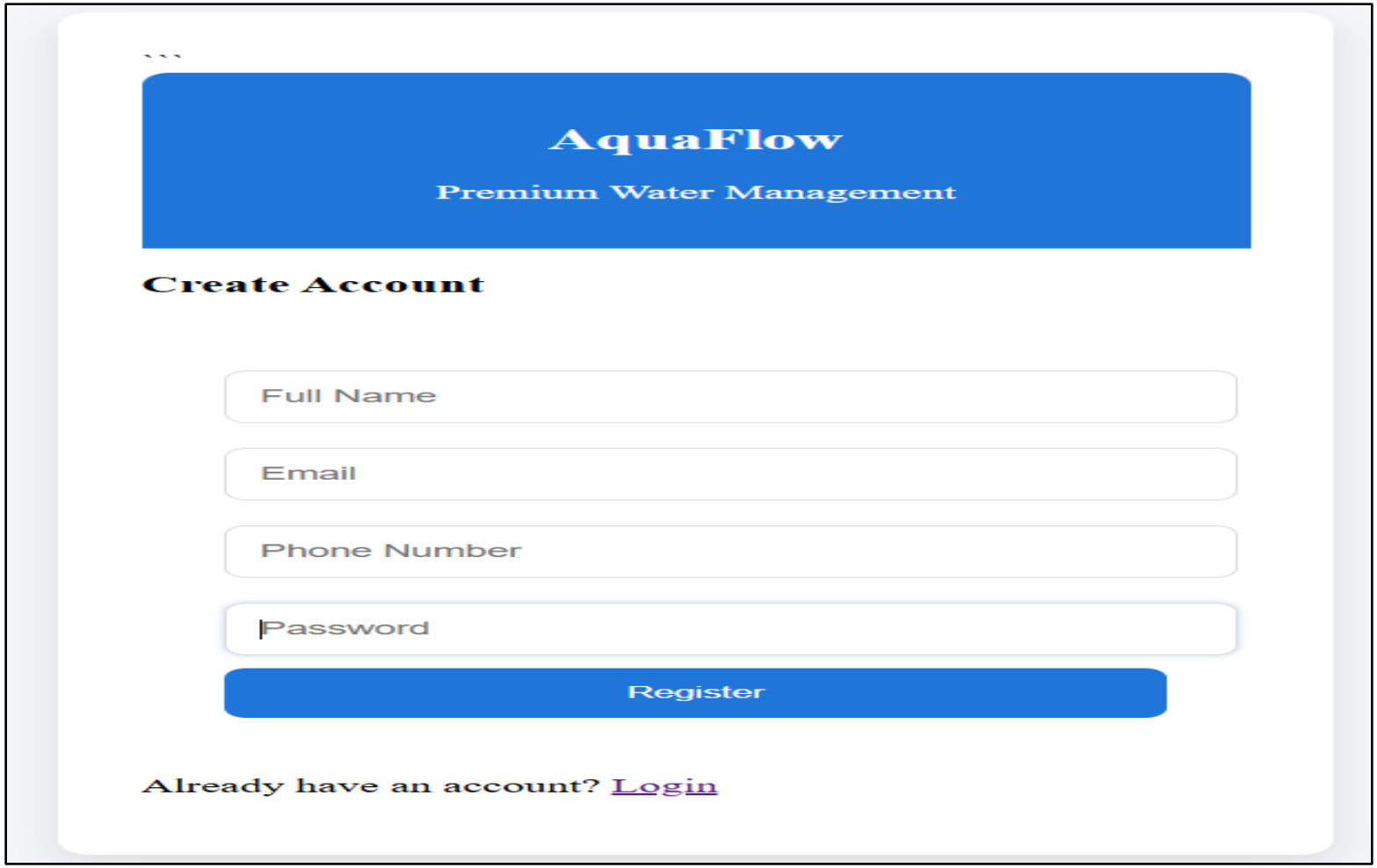
- **About Page**

Why Choose Gurumauli Aqua?
We provide the best water delivery service with quality and reliability.
Pure & Clean Water
Our water goes through rigorous purification processes to ensure the highest quality.
WATER DELIVERY SYSTEM
Fast Delivery
Same-day delivery available. Get your water jars delivered when you need them.
Quality Assured
Regular quality checks and certified processes ensure your safety.
Easy Scheduling
Schedule your deliveries at your convenience. Flexible timing options available.
Jar Return System
Eco-friendly jar return and exchange program. Return empty jars hassle-free.
24/7 Support
Our customer support team is always ready to assist you with any queries.

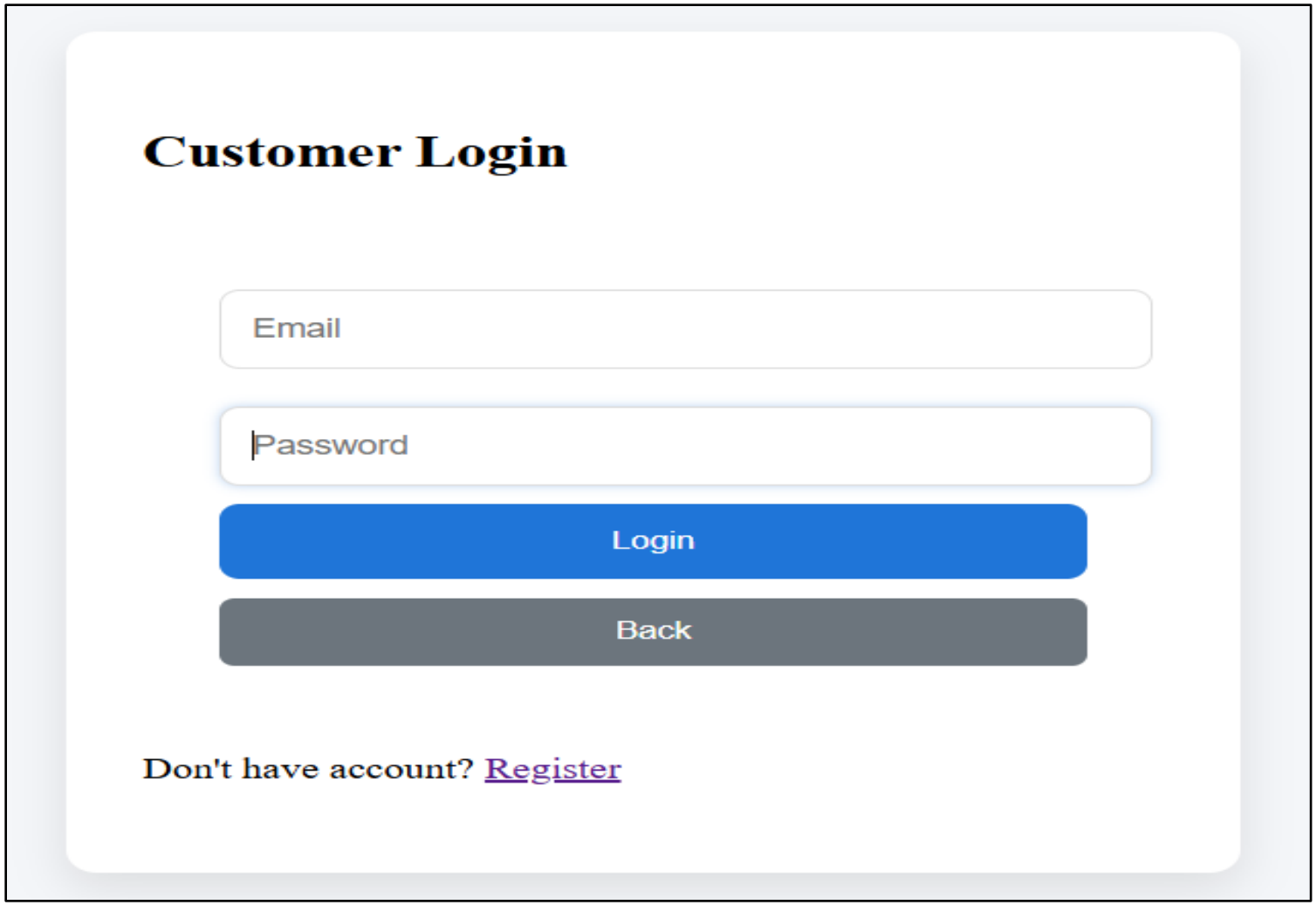
- **Contact Page**



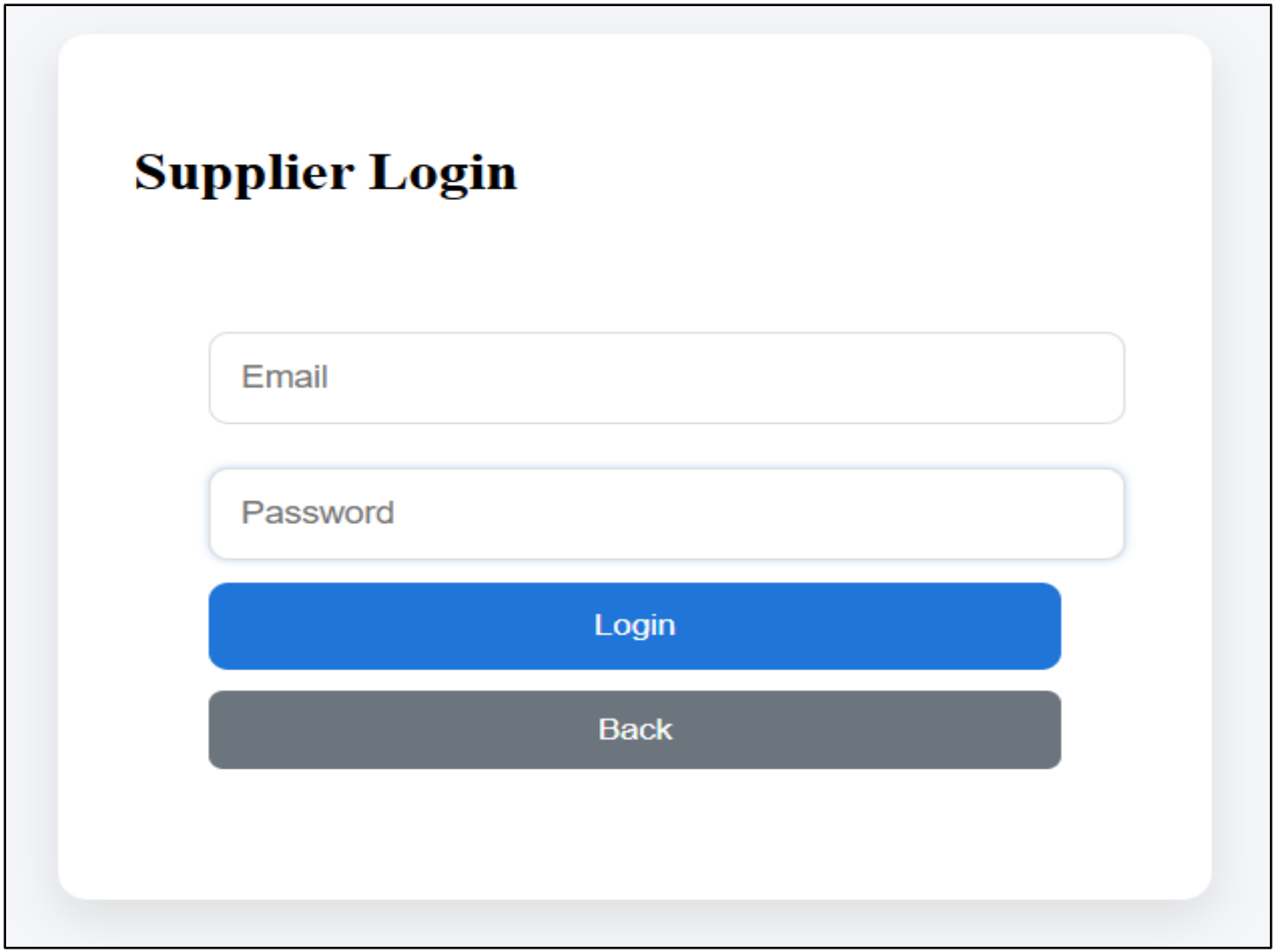
- **Registration Page**



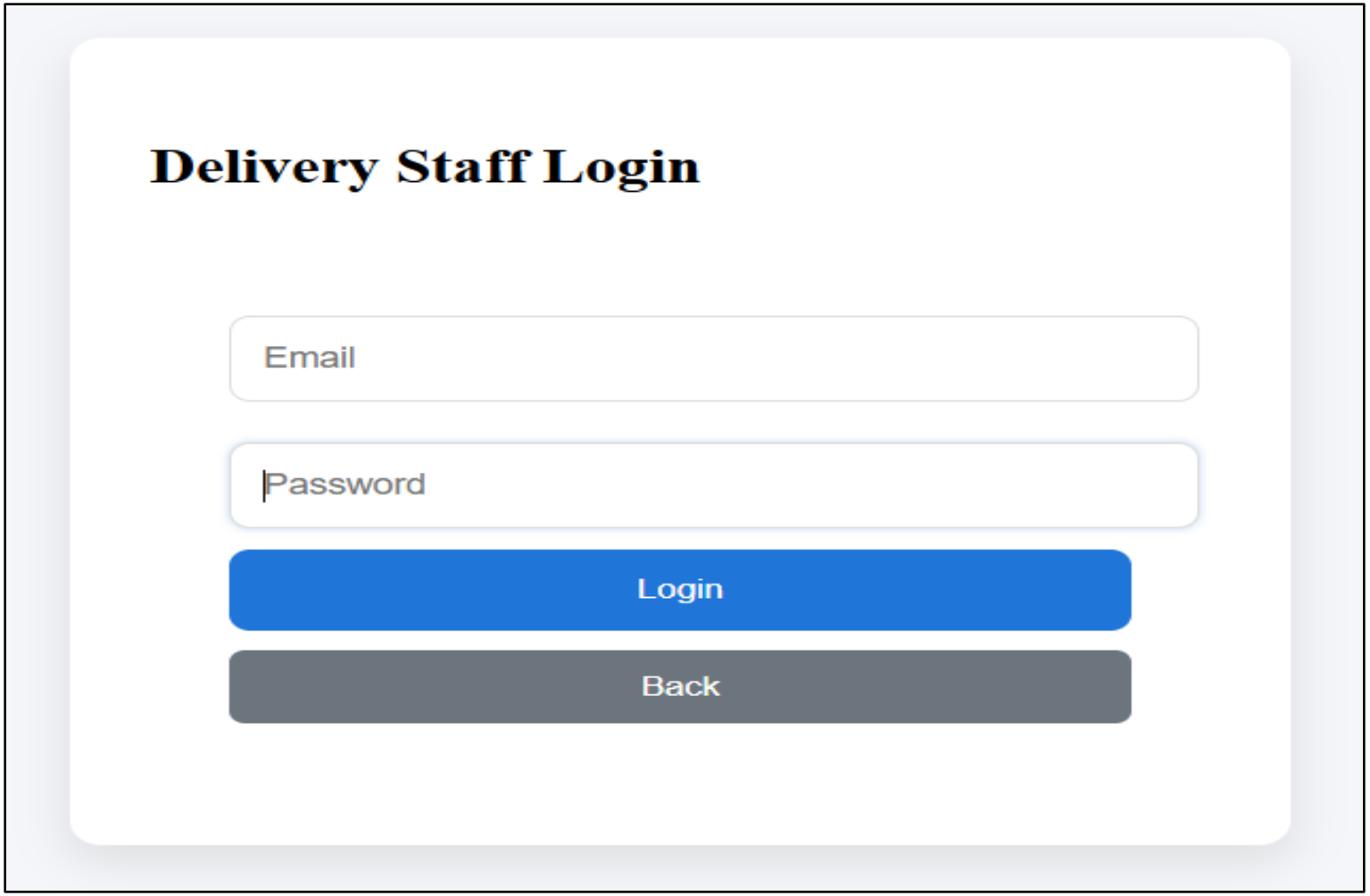
- **Customer Login**



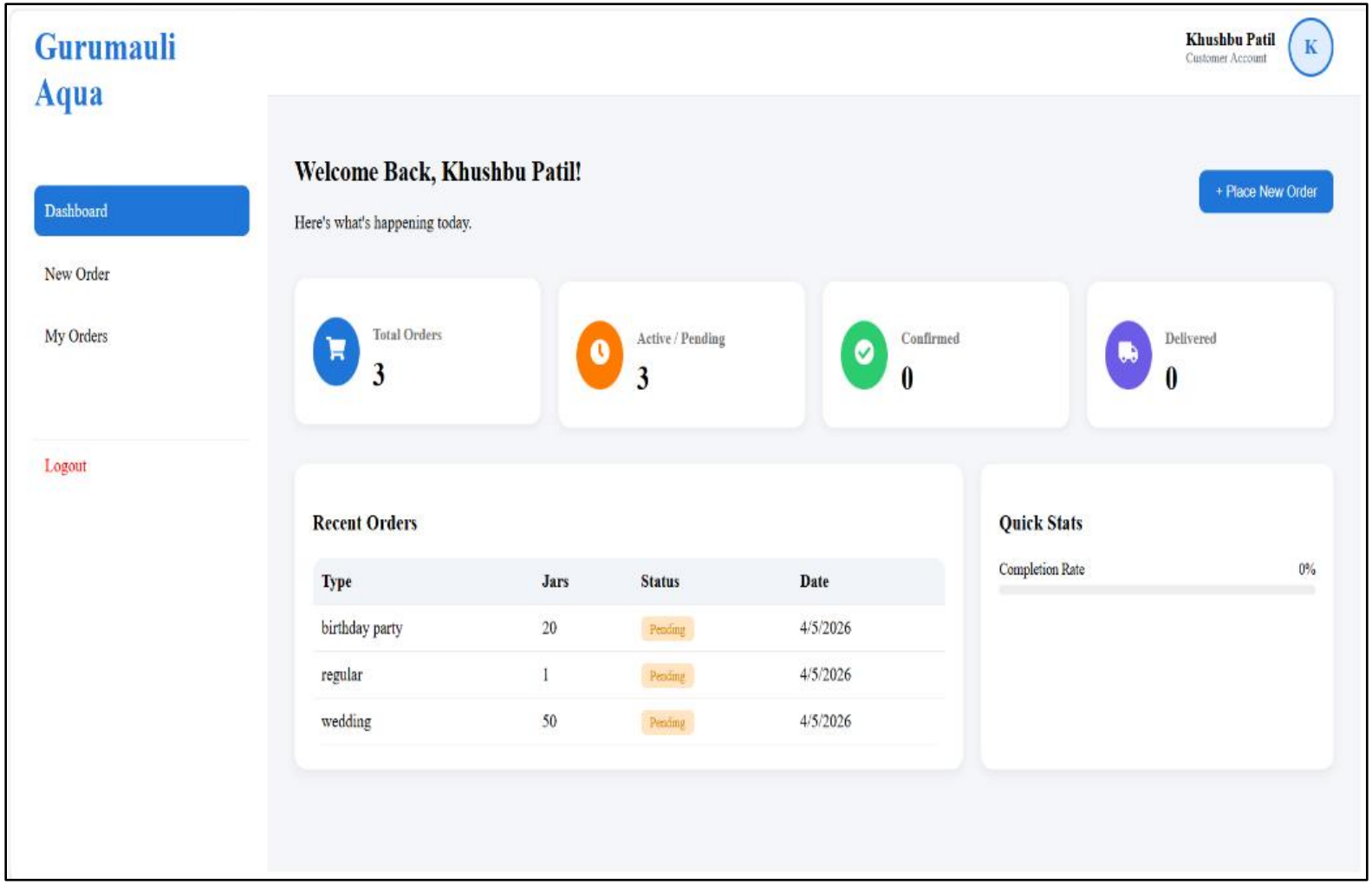
- **Supplier Login**



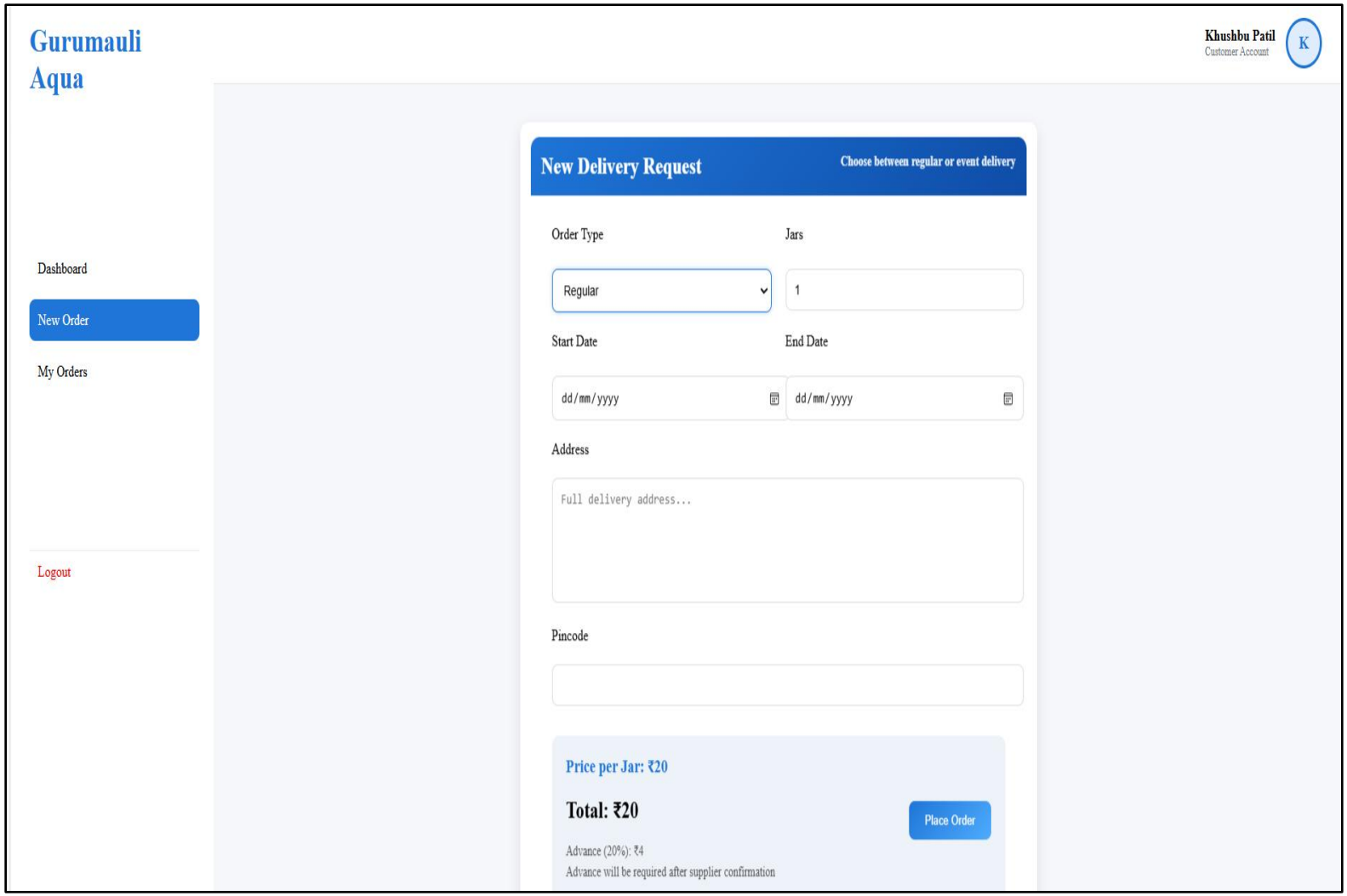
- **Delivery Staff Login**



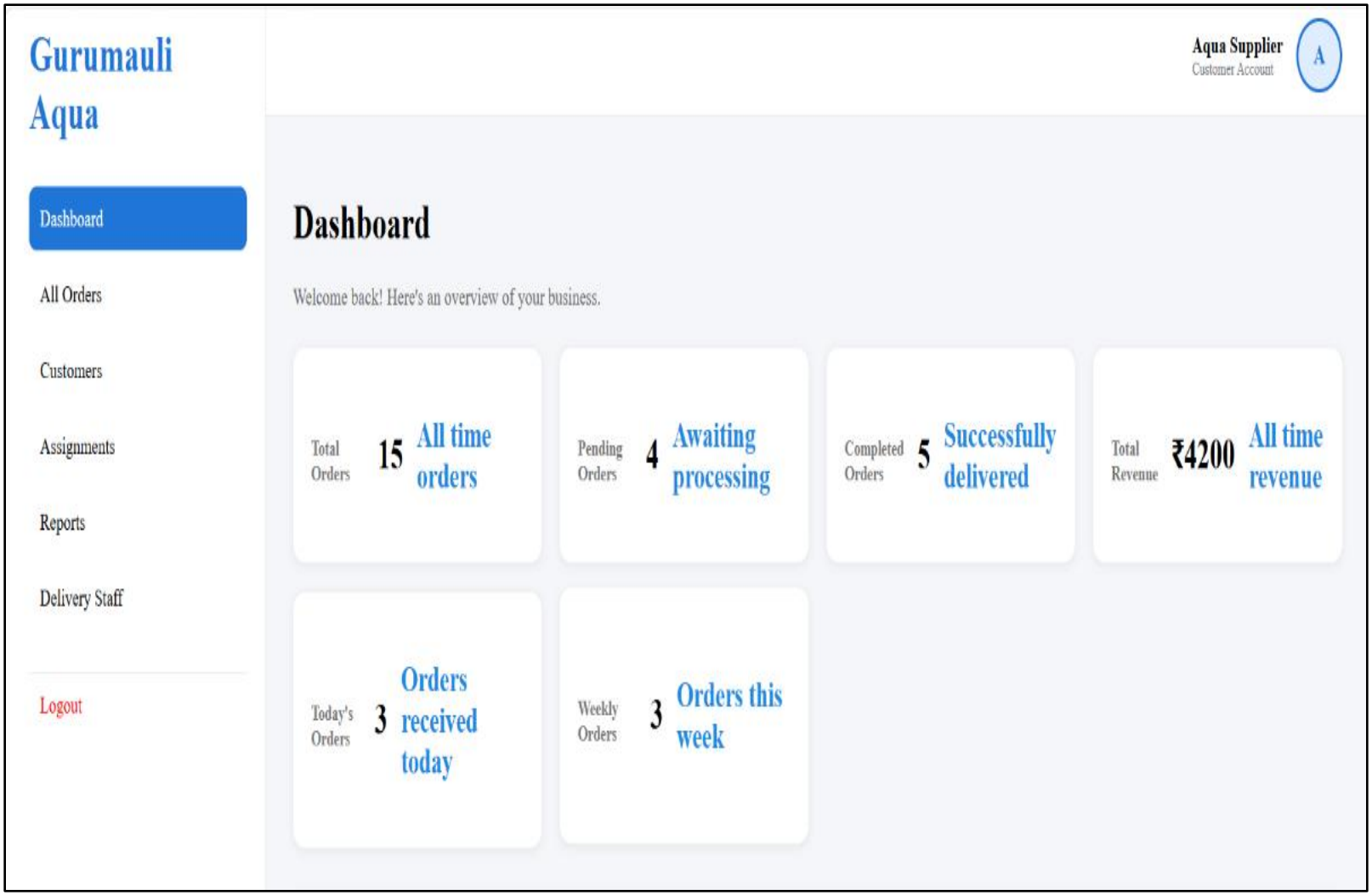
- **Customer Dashbord**



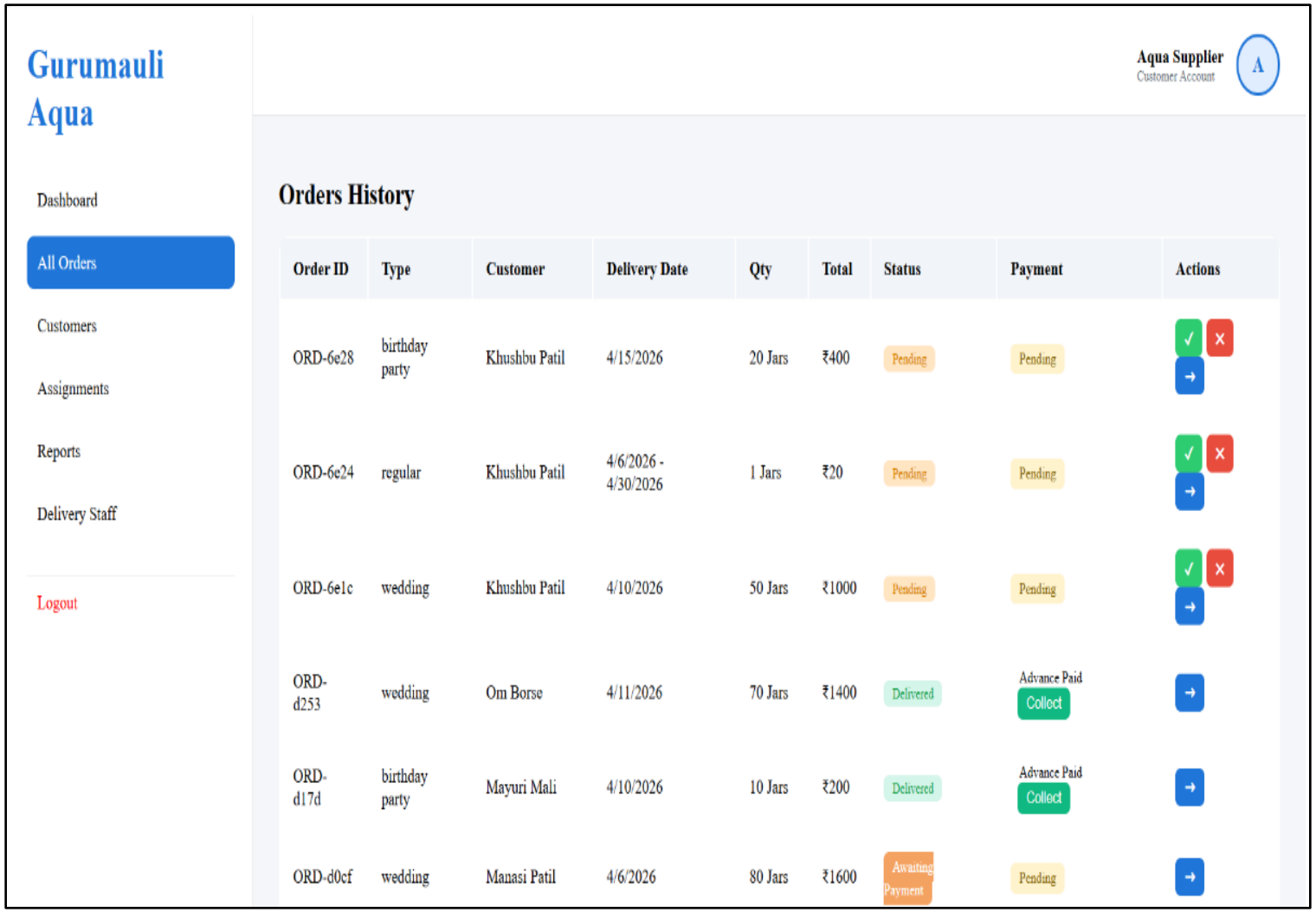
- **New Order Page**



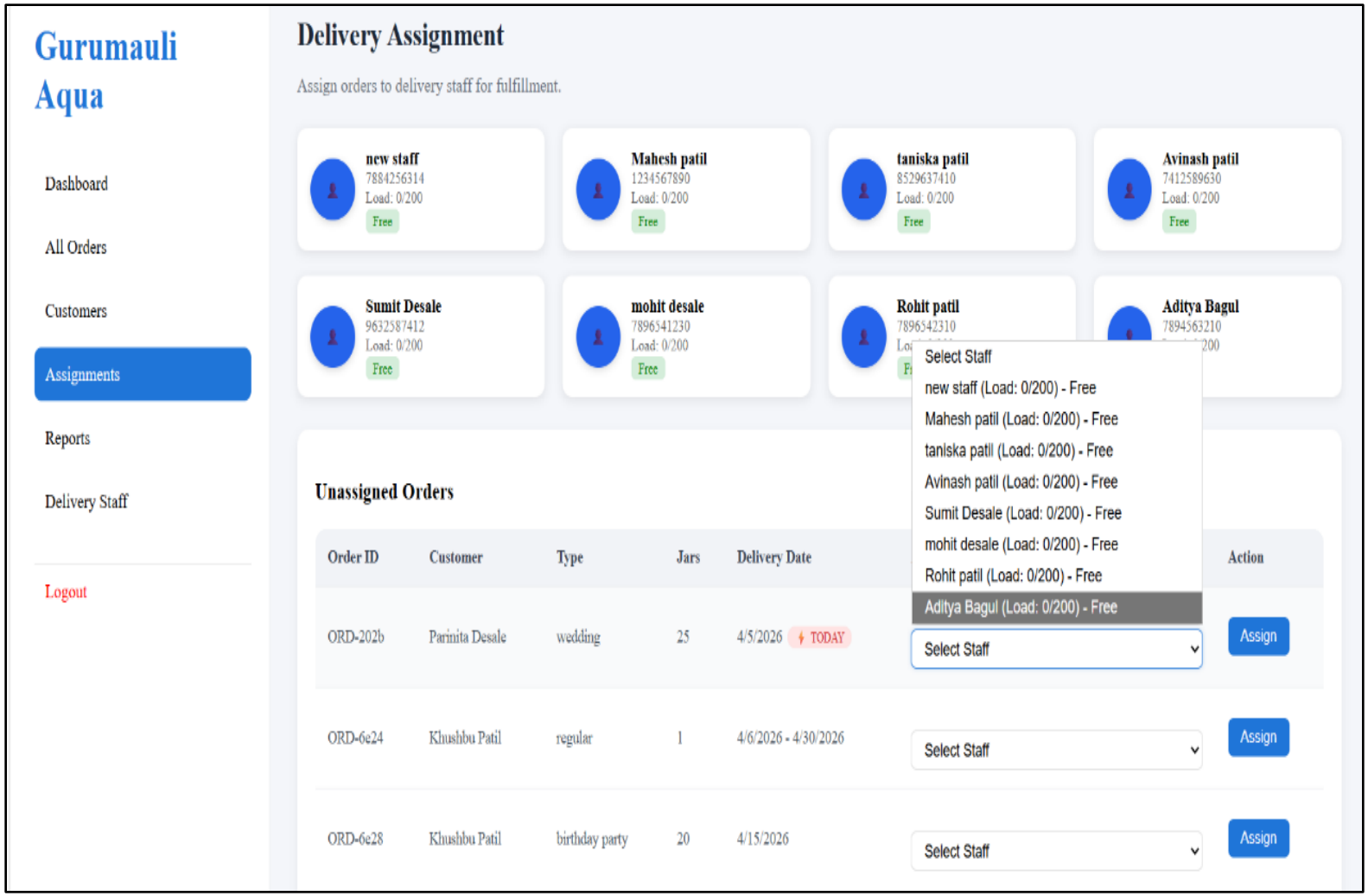
- **Supplier Dashboard**



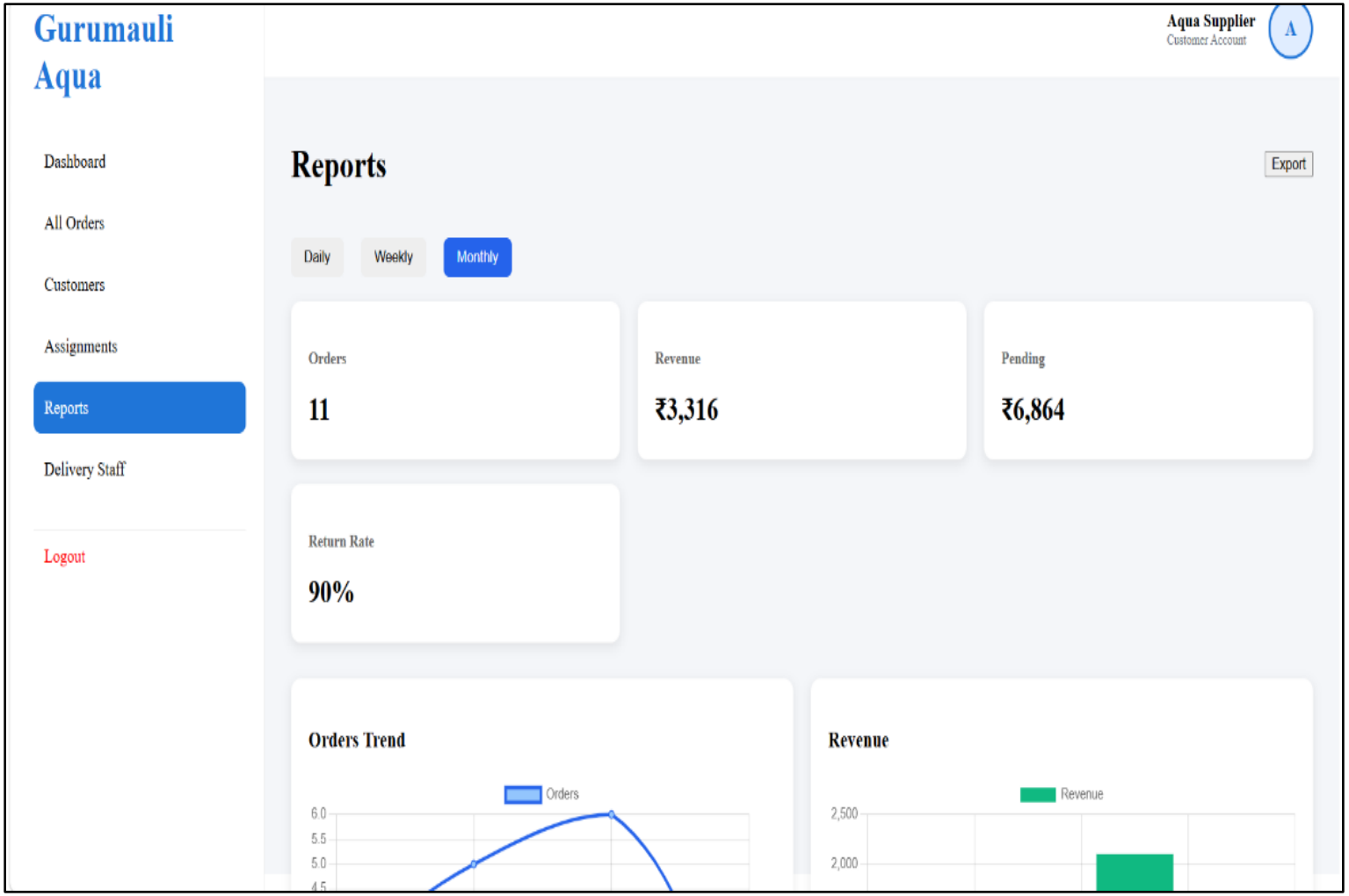
- **All Orders Page**



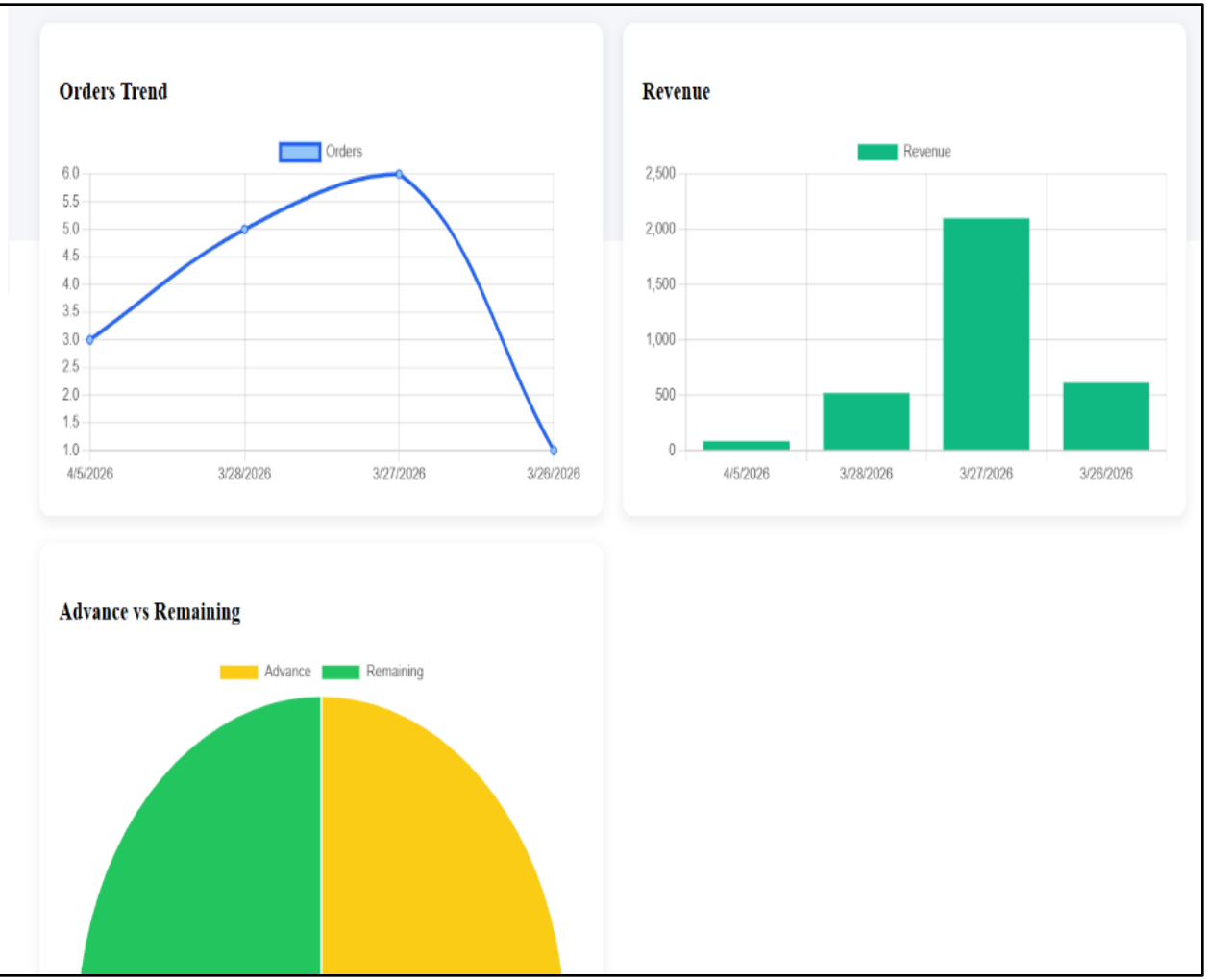
- **Order Assignment Page**



- **Report Page I**



- **Report Page II**



- **Report**

# Gurumauli Aqua

Water Jar Delivery System

## Delivery & Payment Report

Type: MONTHLY

Generated: 4/6/2026, 7:59:30 AM

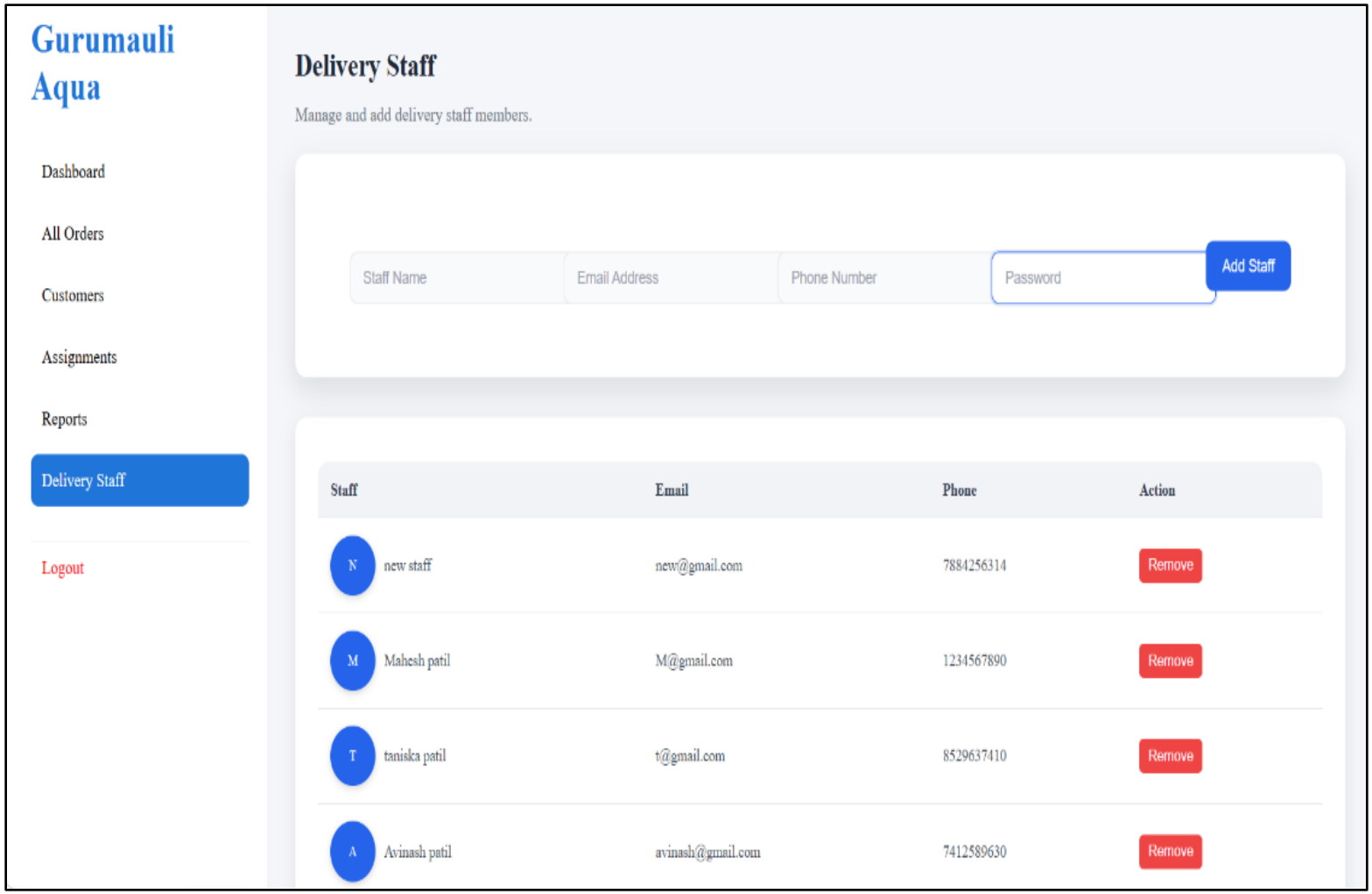
### Fully Paid Deliveries

| Order ID | Customer | Jars | Amount | Status |
|---|---|---|---|---|
| ORD-e620 | Hiteshree Desale | 50 | 1000 | Paid |
| ORD-1fce | Pravin Desale | 30 | 600 | Paid |

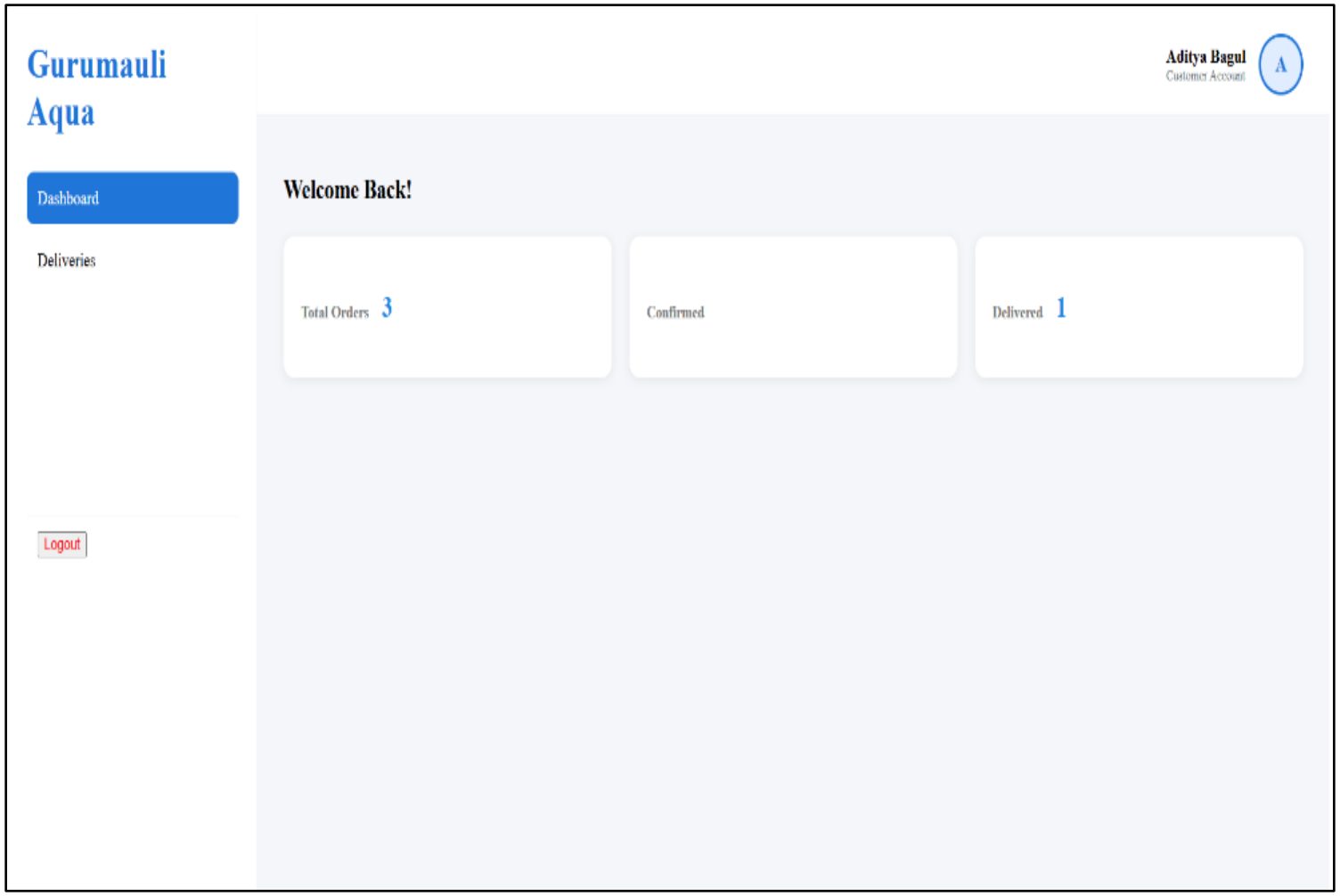
### Pending Deliveries

| Order ID | Customer | Jars | Status | Pending Amount |
|---|---|---|---|---|
| ORD-6e28 | Khushbu Patil | 20 | Confirmed | 320 |
| ORD-6e1c | Khushbu Patil | 50 | Pending | 0 |
| ORD-d0cf | Manasi Patil | 80 | Awaiting Payment | 0 |
| ORD-5075 | Hiteshree Desale | 2 | Pending | 0 |
| ORD-e617 | Hiteshree Desale | 1 | Awaiting Payment | 0 |
| ORD-e611 | Hiteshree Desale | 50 | Pending Delivery | 800 |
| ORD-202b | Parinita Desale | 25 | Pending Delivery | 400 |
| ORD-9afe | ketal patil | 50 | Pending Delivery | 800 |
| ORD-cf3f | Pravin Desale | 153 | Pending Delivery | 2448 |

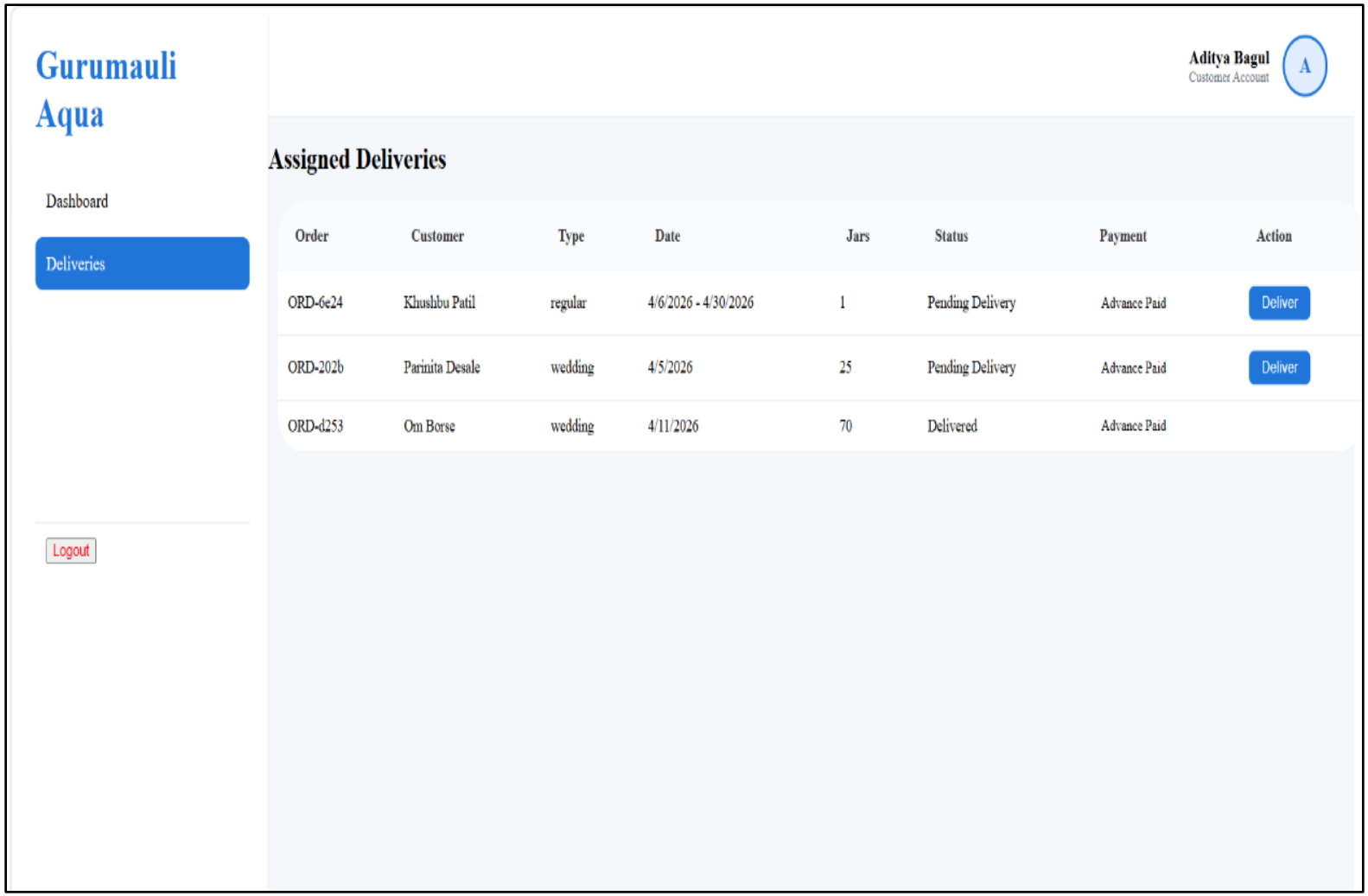
- **Delivery Staff Management Page**



- **Delivery Staff Dashboard**



- **Assigned Deliveries Details Page**



# 7.1 Conclusion

The Web Application for Water Jar Distribution System helps manage water jar orders, deliveries, returns, and payments easily. It reduces manual work and keeps records accurate and organized. This system makes daily operations easier for the supplier and provides practical knowledge of web application development.

# 7.2 Future Scope

- **Online Payment Integration:-** In the future, the system can include online payment options so customers can pay for water jar orders directly through the website.
- **SMS / WhatsApp Notifications:-** The system can send SMS or WhatsApp notifications to customers for order confirmation, delivery updates, and payment reminders.
- **Mobile Application:-** A mobile application can be developed so customers and suppliers can access the system easily through smartphones.
- **GPS-Based Delivery Tracking:-** GPS tracking can be added to track delivery locations and ensure timely delivery of water jars.

# 8. References and Bibliography


- JavaScript Documentation
- React JS Official Documentation
- Node.js and Express.js Documentation
- MongoDB Official Documentation
- www.w3schools.com
- www.geeksforgeeks.org